

**The AI Dilemma: A Behavioral Economic Analysis of AI Incentives in Higher Education**

Ruth Feleke and Mary Kreklow

Department of Economics, St. Olaf College

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Dr. Ashley Hodgson

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## Introduction

The rise of Generative AI in higher education has created a classic Tragedy of the Commons problem where the private utility gained by an individual student through automated assignment completion threatens the collective value and integrity of a university degree. When students use AI to finish assignments instantly, they bypass the struggle necessary for real learning. Academic rigor is a public good; however as students increasingly delegate cognitive labor to AI, they participate in a systemic atrophy of critical thinking. Students often turn to AI because the pressure to perform well tends to outweigh the incentive to learn. To protect academic integrity, and support students in attaining genuine learning, institutions must move beyond simple detection and instead change the incentives that make AI so attractive. This paper analyzes two different structural policies designed to change the choice architecture and make AI usage less appealing for students.

The Submission Stages policy changes how assignments are submitted. Instead of only requiring a final submission, students would be required to submit a paper trail of their work, including outlines, rough drafts and edit histories. This removes the primary benefit of AI, which is efficiency. If a student has to show how their ideas evolved over time, they can no longer use AI to generate the perfect essay or answer in seconds. This approach also makes it easier and more cost effective for teachers to detect academic dishonesty by allowing them to see whether a student's final work matches their early drafts. By breaking work into smaller steps, this policy also helps reduce the stress that often leads students to use generative AI in the first place.

The Random Oral Exam Policy focuses on verification and consequences. To ensure that a student actually wrote their work, 10% of all assignments will be randomly selected for an in-person oral exam. This forces students to understand their own submissions as they might have to defend their ideas at any given time. If a student is caught using AI the consequences range from receiving a grade reduction or resubmitting the assignment to having a permanent mark of academic dishonesty placed on their transcript. This policy aims to lower the utility of AI by making it a high-risk, and high effort way to cheat, eventually nudging students back toward completing assignments without the use of generative AI.

However, implementing these two policies would create a significant challenge for students who use AI to enhance, rather than replace, their own work. If students use AI to explain complex concepts, AI would be a benefit to genuine learning. In reality, the ambiguous line between academic dishonesty and the legitimate use of AI as a resource makes it difficult to determine whether students are strictly enhancing their learning or bypassing cognitive labor. To maintain a focused analysis, this paper specifically addresses instances in which students use AI as a substitute to complete their work rather than as a tool to augment it. Furthermore, these policies create an additional burden for faculty. While administrators prioritize the long-term value of the degree, individual professors face a hefty increase in their daily workload. Without additional institutional support, these high costs may lead to inconsistent enforcement, leaving the same gaps in information that allowed AI misuse to flourish in the first place.

This research applies several core behavioral economics concepts to explain the shift in student decision making. Specifically, this paper explores how present bias and hyperbolic discounting nudge students to overvalue the immediate efficiency of AI, even when it conflicts with their long term goal of intellectual mastery. In addition, this paper examines how information asymmetry creates an environment in which students can prioritize automated assignment completion over genuine learning, simply because the institution lacks the framework to verify authentic cognitive labor. By evaluating the practical implementations of these policies, this paper will demonstrate how redesigning academic incentives ensures that students can foster strong critical thinking skills and mastery of their education.

## **Literature Review**

Recent studies suggest that if students continue to prioritize AI driven shortcuts over the mastery of underlying theories, the result will be a “future shortage of specialists” because high level expertise requires an irreplaceable human element (Schei et al., 2024). This creates a profound Principal-Supervisor-Agent conflict between university administrations and students. While institutional policies are often optimistic about AI as a productivity tool, a number of researchers argue this stance is a subpar approach to address plagiarism and ultimately weakens students' intellectual growth as well as compromises their long term autonomy. (Abbas et al., 2025; McDonald et al., 2025)

At the individual level, the decision to utilize GenAI (Generative AI) is a classic problem of intertemporal choice characterized by significant present bias. Students frequently overvalue the immediate convenience and minimal effort that using AI to complete assignments allows. As a result, the delayed and more abstract utility of deep intellectual development gets left behind. This bias is triggered by environmental stressors. Research indicates that a high academic workload and time pressure act as primary catalysts nudging students towards AI reliance as a survival strategy (Abbas et al., 2024). To explain this behavior, researchers point to the Technology Acceptance Model (TAM), which suggests that perceived usefulness and perceived ease of use are the primary incentives for AI usage amongst students. The perceived usefulness of AI is so high that it incentivizes a “surface level approach to learning,” where students only have to invest the bare minimum mental effort to fulfill the requirements of the assignments (Schei et al., 2024). The immediate reward of task completion masks the long-term consequences of AI overreliance such as increased procrastination, memory loss and a lower cumulative GPA (Abbas et al., 2024).

The struggle between immediate task completion and long term skill acquisition is rooted in hyperbolic discounting, a behavioral model in which the perceived value of a reward drops significantly as the delay to receive it increases. Students reflect present bias by prioritizing the relief of submitting an assignment and the reclamation of free time over the delayed payoff of critical thinking skills and building expertise. Since the costs of AI overreliance are abstract and temporarily distant, they are heavily discounted in the student’s current decision making process. As a result, using GenAI to complete assignments becomes the rational choice for students, even when they logically understand that this choice undermines their long term intellectual development.

This behavioral shift is further influenced by changing social norms and peer pressure. According to the Unified Theory of Acceptance and Use of Technology (UTAUT), which is a theoretical model used to explain technology adoption, social influence makes AI usage all the more appealing. As GenAI is further incorporated into the classroom, the socially optimal behavior (independent work requiring critical thinking) becomes more expensive in terms of relative time and effort. If a student perceives that their

peers are achieving similar grades with less effort, the fear of falling behind creates a race to the bottom in cognitive engagement. Some researchers propose the AI-ICE (Ideas, Connections, Extensions) model, which is a teaching framework that breaks learning into the three stages to guide students towards using AI as a starting point while keeping the critical thinking and final conclusions for themselves. However, the current norm favors an AI-directed rather than an AI-supported paradigm. Without a structural intervention in the choice architecture of higher education, the default state remains a reliance on a tool that provides a degree of convenience and efficiency at the expense of the deeper understanding that is required for long term intellectual development.

One of the largest challenges surrounding AI policies is information asymmetry (Azevedo et al., 2026). Current existing policies are often ambiguous, leaving students with limited knowledge of expectations and enforcement. Furthermore, AI use is not easily observed; in fact, teachers are often overconfident in their detection abilities and are not able to effectively differentiate between AI generated and student-written essays (Fleckenstein et al., 2024). Because educators are not able to tell when an assignment is human-created, they are forced to treat all assignments equally when evaluating. Educators often prefer AI generated work when judged blindly (Fleckenstein et al., 2024), making submitting AI generated work the optimal choice for students hoping to receive a good grade. Additionally, most existing policies leave regulation up to individual instructors, leading to varying guidelines and continued ambiguity for students as to what is considered acceptable (Azevedo et al., 2026). As a result, students often underperceive consequences, as guidelines are not in place. Implementing the random oral exams policy closes this information gap by having clear guidelines and making everyone aware of the uniform process. The guidelines are implemented at the university level, ensuring that students, teachers, and administrators alike have the same information about policies, expectations, and enforcement.

The lack of clarity around AI policies leads to moral hazard for students. When consequences are either unclear or delayed, students will be more likely to take the risk of using AI because of self-serving bias, which allows them to interpret the ambiguity in a way that justifies behaviors that they might otherwise view as dishonest. This outweighs the effort reduced by not doing the work by hand and

underweighs potential consequences, including violations and a lack of genuine learning (Wang et al., 2025). Random oral exams can help with this by working to increase students' genuine learning and putting higher salience on the costs of using AI. In doing so, students become less likely to submit AI generated work and are led to prioritize longer-term learning.

AI has three main uses in education. Task displacement involves AI doing a task instead of the student, risking skill erosion and leading to lost practice opportunities. Task augmentation occurs when AI supports the task without displacing the student's cognitive effort. This allows enhanced opportunities for the student to practice. Task reinstatement is when AI enables students to complete a previously not feasible task, which opens up new practice opportunities (Chirikov, 2026). This also aligns with substitution and complementation. Depending on its use, AI can act as a substitute for learning, where critical thinking and genuine learning decline in students, or it can act as a complement, allowing students to learn more and increase their critical thinking and genuine learning. Due to these varied effects, policies should also be differentiated to discourage displacement while encouraging augmentation and reinstatement. With both submission stages and random oral exams, students are not necessarily prohibited from using AI; rather, they aim to discourage overreliance and specifically target displacement. The use of AI as augmentation or reinstatement is not directly prohibited, so guidance on AI use is left to the professor's discretion.

To effectively nudge students away from AI misuse, educators must identify the underlying motives leading to reliance on the tool. Oftentimes, assignment difficulty or lack of perceived support can lead to higher use rates of AI (Qin, 2025), as students feel unprepared to complete the work. Redesigning assignments can help reduce this reliance. Changing formats of submissions can also discourage use of AI by requiring students to give a short oral defense of a paper, proctoring writing, or using technology that detects key strokes for submitted essays (Taylor, 2026). The submission stages policy reformats how the assignment is broken down with the goal of decreasing assignment difficulty to ensure students feel they are set up to complete the assignment on their own.

Students are often nudged by social norms, not just rules, so policies should consider social aspects alongside regulations. Submitting AI generated work can be equated to plagiarism, as it involves submitting work not written by the student, and credit is not given to the actual creator. All educational institutions restrict plagiarism, and AI could be regulated in the same manner (Taylor, 2026). One student using AI may not be a huge deal, but it becomes a tragedy of the commons issue when that human created work is being compared to AI generated work. It has been shown that educators often prefer AI generated writing over student writing (Fleckenstein et al., 2024). This holds particularly true with written assignments such as essays, but may not hold for more creative assignments showcasing more human qualities of students. Regardless, for many assignments the standard is raised by one student submitting an AI generated submission. Others will begin using AI in an attempt to keep up, devaluing the degree from that institution and falsely testifying to the ability of a student (Taylor, 2026). By discouraging AI misuse and shifting the optimal amount of AI generated work submitted down, the norm becomes turning in self-completed work, the standard is set to students' actual writing level, and the pressure to use AI to keep up to an AI standard will decrease, further lowering incentives to submit AI generated work.

### Game Theory

The following matrix illustrates the interaction between a representative Student (Player 1) and the University Administration (Player 2). Payoffs are represented as (*Student, Administration*).

|                                       | <u>No Policy</u> | <u>Implement Policy</u> |
|---------------------------------------|------------------|-------------------------|
| <u>Submit AI generated work</u>       | <u>4</u> , 1     | 1, <u>2</u>             |
| <u>Don't submit AI generated work</u> | 2, <u>4</u>      | <u>3</u> , 3            |

*Table 1:* Game theory matrix of dynamics around submission of AI generated work and policies between students and administration.

The interaction between students and university administration can be modeled in a non-cooperative game. The payoff matrix reflects the distinctive differences in priority between students and administration. To understand the strategic choices of both players, we must define the utility values

driving these numbers. The student receives the highest utility (4) when they submit AI generated work without facing policy restrictions because this maximizes efficiency and minimizes stress. Conversely, the student receives the lowest utility (1) when they submit AI work under a strict policy, under which high detection risks and transcript penalties outweigh the time saved. The administration reaches its highest utility (4) when students work independently without the school needing to fund expensive enforcement. The administration reaches its lowest utility (1) if students use AI unchecked, as this negatively impacts institutional credibility and reduces students' genuine learning.

The misalignment between students and university administration prevents the game from reaching a stable Nash Equilibrium. If the administration maintains a “No Policy” stance, the student logically defects from submitting independently completed work to submitting AI generated work to increase their payoff from 2 to 4. This shift forces the administration's payoff down to 1, prompting them to move to the “Implement Policy” stance to achieve a higher payoff of 2. Once the administration enforces these rules, the student finds that independent work offers a higher utility of 3 compared to the high-risk payoff of 1 for cheating. However, as student compliance increases, the administration views the high labor costs of tracking drafts and conducting oral exams as an inefficient use of resources. To mitigate these operational costs, the administration reverts to the “No Policy” state to secure a payoff of 4. This creates a perpetual cycle of enforcement and relaxation of policies where both actors continuously adjust their strategies in response to the other's incentives. To find an equilibrium, the university must restructure the choice architecture to ensure that independent work remains the students optimal strategy.

A critical challenge to this equilibrium is the potential for students to “trick” tracking progression by using AI to stimulate multiple drafts, or artificial typing patterns. However, the Submission Stages Policy remains effective because it increases the cost of cheating regardless of how sophisticated AI's technology is. Even if a student uses AI to generate multiple drafts of an assignment, they are forced to continuously engage with the material. Students will need to review, organize and submit each incremental step, which prevents the low effort payoff that makes AI attractive in the No Policy state.

Furthermore, the policy restructures the students' utility by introducing psychological costs. By requiring frequent draft submissions and random oral exams, the administration makes the interactions more personal. This creates shame for the student, especially when considering the mental effort to maintain a lie over multiple stages. While hyperbolic discounting usually allows students to overvalue the immediate relief using AI to complete assignments while leading the risk of detection to their future selves, these policies force those future consequences into the present. By including frequent checkpoints and random oral exams into the curriculum, the administration creates an immediate cost for dishonesty through social shame and the pressure of being questioned in person. When the effort needed to successfully trick the multi-stage system meets the effort required to complete work independently, the students' rational choice shifts towards genuine learning, finally stabilizing the equilibrium.

### **Model and first order condition interpretation**

#### School Administration:

$$\max_{B,R} D(A(e(B,R))) + M(A(e(B,R))) - F(e(B,R)) - U(e(B,R))$$

Where:

- B: Submission Stages policy
- R: Random Oral Exams policy
- e: Level of effort [0,1] put into implementing policies by professors
- D: integrity/reputation of college degree
- A: % of AI generated work submitted
- M: mastery of knowledge/critical thinking in students
- F: time burden and policy enforcement costs
- U: student autonomy

#### Professor:

$$\max_e \alpha L(A(e(B,R)) - T(e,B,R,O,P)$$

Where:

- e: Level of effort [0,1] put into implementing policies
- L: Students' genuine learning
- $\alpha$ : Care for students' genuine learning (assuming  $\alpha > 0$ )
- T: Opportunity cost of time and effort required to implement policies
- R: Random Oral Exams policy

- B: Submission Stages policy
- A: % of AI generated work submitted by students
- O: Other responsibilities held by professor
- P: Support (TAs/Graders) available to assist professor with workload

Student:

$$\max_A -S(A, e(B)) + E(A, e(B)) - V(A, e(R)) + \beta L(A, e(R, B))$$


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Where:

- A: % of AI generated work submitted
- S: Stress of school work
- E: Efficiency of completing schoolwork
- V: Expected Violation Cost
- $\beta$ : Desire for Genuine Learning (assuming  $\beta > 0$ )
- L: Genuine Learning
- R: Random Oral Exams policy
- B: Submission Stages policy

We utilize a principal-supervisor-agent framework to capture misaligned motives around submission of AI generated work between students and administration. Through this, we explore incentives that encourage students to submit AI generated work, incentives pushing professors to actually spend time and effort implementing policies, and incentives that lead administrators to impose policies to reduce misuse of AI. This framework builds on research showing that information asymmetry and ambiguity around policies lead to higher rates of AI submitted work (Azevedo et al., 2026). School administrators face a tradeoff between policy enforcement and cost, as the measures taken to reduce AI submissions are expensive and time consuming (Fleckenstein et al., 2024). They also risk harming relationships with students, as policies may decrease student autonomy. Both the submission stages policy and the random oral exams policy lead to extra time and energy burdens on educators, as extra time must be spent either grading and checking submissions or evaluating oral defenses. This burden is reflected in the professor model, as professors directly take on the burden and therefore must decide how much effort to put into the implementation of the administration's chosen policies. Professors work to balance their students' genuine learning while considering the opportunity cost and tradeoffs they may be forced to

make, depending on workload and support, to dedicate time and energy into enforcing policies. The policy chosen by administration and effort professors put into enforcing it will factor into students' decisions regarding the amount of AI generated work to submit.

Students are pushed to prioritize efficiency due to present bias and often rely on AI in times of high workload and time pressure (Wang et al., 2025; Abbas et al., 2024). They strive to finish work as quickly as possible without compromising their grade. This highlights internal motives behind AI use. However, they also often report concerns about AI, particularly commenting on the risk of losing creativity and learning and worrying about consequences (Baek et al., 2024). This model allows us to incorporate findings from the literature into the relationship between administrators and students, evaluate how our policies can change incentives, and explore the ultimate effects on behavior. With the utility maximization framework, we can identify which incentives must change to reduce AI misuse.

The first order condition for the student is:

$$\frac{\partial \pi}{\partial A} : -\frac{\partial S(e(B))}{\partial A} + \frac{\partial E(e(B))}{\partial A} - \frac{\partial V(e(R))}{\partial A} + \frac{\partial \beta L(e(R, B))}{\partial A} = 0$$

We can rewrite this as:

$$-\frac{\partial S(e(B))}{\partial A} + \frac{\partial E(e(B))}{\partial A} = \frac{\partial V(e(R))}{\partial A} - \frac{\partial \beta L(e(R, B))}{\partial A}$$

Administrators choose policies (submission stages or random oral exams) by anticipating students' optimal amount of submitted AI generated work. Weak policies are shown to lead to misuse of AI, while overly strong enforcement creates a large burden and resistance (Azevedo et al., 2026), as well as loss of students' perceived autonomy. Therefore, administrators must work to maintain their degree integrity and learning outcomes while balancing costs. The optimal amount of AI generated work submitted by students is influenced by present bias, social norms, moral hazard, and personal desires for learning. Overuse and misuse of AI often emerges as a response to these existing incentives.

A common justification for promoting the integration of AI use in education is that, when used complementarily, it can increase thinking and learning (Chirikov, 2026). This is reflected in our model

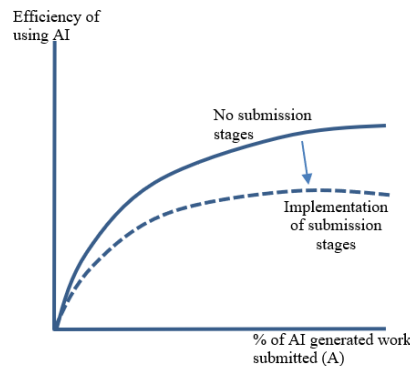
through increased efficiency and decreased stress with higher proportions of AI generated work submitted. This perspective would therefore support not disincentivizing AI use. However, if the convenience of submitting AI generated work dominates learning, AI becomes a substitute to actual learning, thereby reducing the integrity of the degree. By including genuine learning in the student's model and critical thinking and degree integrity in the administrator's model, we capture the downsides of reliance on AI and illustrate the tragedy of the commons that is created by unrestricted AI usage. While individual students may benefit from gained efficiency and stress reduction, collective overuse will reduce overall learning and can undermine the value of a degree.

To discourage misuse of AI, administrators may consider requiring multiple stage submissions for assignments, as outlined by the Submission Stages policy. By splitting up the work, overall stress is reduced, as students are faced with smaller portions of work at a time. This stress reduction is expected to have a high elasticity effect on the amount of AI generated work submitted, as the link between stress and AI use has been shown to be strong (Abbas et al., 2024). The increased efficiency of schoolwork achieved through using AI is also decreased while genuine learning is increased because submission stages allow for more focus on the process of the work, and less focus on the end product, which could quickly be generated by AI (Qin, 2025; Chirikov, 2026). Because of this, the marginal benefit of submitting AI work is decreased and the marginal benefit of genuine learning is increased.

The administration may also consider implementing random oral exams for assignments. This increases the expected cost of violation (Azevedo et al., 2026), as students will be randomly assessed on their understanding of an assignment in real time. Being able to tell when a student truly understands the material is much easier with an oral format, as it forces the student to demonstrate real-time synthesis rather than just outputting a result. It also leads to higher genuine learning, as knowing the material is necessary to do well on an oral exam. This increases the marginal cost of AI use, shifting the equilibrium to lower levels of AI generated work submitted by students.

### **Comparative Statics**

*Elasticity of amount of AI submitted work w.r.t efficiency of submitting AI work over completing independently.*



*Figure 1: Impact of submission stages policy on student responsiveness to AI efficiency gains.*

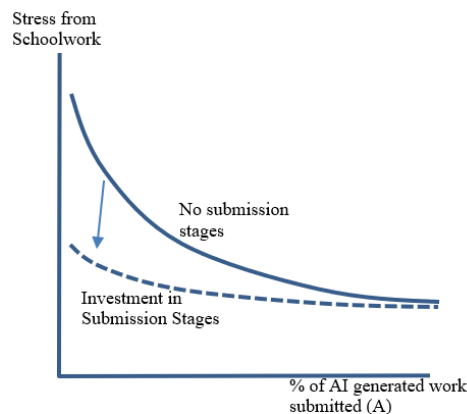
The model begins at the (0,0) intercept, representing a baseline state where the student submits zero AI generated work and consequently gains zero efficiency from using AI to complete work. As the student begins to delegate tasks, the curve rises steeply, reflecting high initial elasticity. At this stage, even a small increase in AI usage generates substantial efficiency gains by relieving the mental burden of brainstorming and outlining. The curve adopts a concave shape to illustrate diminishing marginal utility. As the percentage of AI generated work increases, each additional increase in AI usage yields smaller efficiency returns because the student has already offloaded the most time-intensive portions of the assignment. The student's sensitivity to this efficiency gap dictates the slope of the curve. During high stress periods like finals week, the students' preference for efficiency intensifies, causing the curve to steepen as the relative cost of independent work rises.

The submission stages policy directly intervenes in this choice architecture by shifting the entire curve downward. By requiring drafts and editing histories, the university administration imposes an additional labor burden on students who attempt to use AI. This policy forces students to manually construct a work history, which effectively removes the time saving advantage of generative AI. Consequently, the downward shift of the curve indicates that AI no longer provides a rational shortcut, as the effort required to forge a believable paper trail equals or exceeds the efforts of genuine learning. This

structural change ensures that even when efficiency is the students primary goal, independent work remains the more viable option.

### *Elasticity of AI submission w.r.t. stress*

Research links workload and time pressure, two signs of stress, to increased reliance on AI (Abbas et al., 2024), suggesting that the marginal benefits of AI rise sharply when students are experiencing high stress.



*Figure 2: Student responsiveness to academic stressors and the moderating effect of the submission stages policy*

As shown in Figure 2, stress levels are highest with no AI use, as this means the student is taking on the entire workload on their own. They will first offload the most stressful tasks to AI, and once those are off their plate, each additional task will decrease stress a bit less than the last. The concavity illustrates the diminishing marginal returns.

If the elasticity of AI submitted work with respect to stress is high, this implies that even small increases in stress result in large increases in AI use. Submission stages decrease stress for all students, regardless of current amount of AI generated work, as the assignment is now broken into much more manageable submissions. As seen through present bias, a student with an assignment due in the future is likely to push back worry until the last moment. This behavior is driven by hyperbolic discounting, where the perceived cost of completing the work feels negligible when the deadline is distant, but the immediate burden of effort is avoided at all costs. This leads to large amounts of stress and workload right before the

assignment is due, as they struggle to complete the entire thing at the last minute. With the submission stages, there will now be smaller, incremental checkpoints throughout the timeline of the assignment, which breaks it down into smaller pieces and is much more tangible for students. The submission stages policy provides the largest reduction in stress from schoolwork for students who are already not using AI, as they are currently not offloading their stress onto AI. They will see the largest drop in stress, as submitting version histories takes little more than a few clicks, and the reduction from breaking down the assignment outweighs the potential stress or added workload of submitting version histories. As the percent of AI generated work increases, the reduction in stress from the burden of schoolwork becomes smaller; however, the flattened curve largely dampens the high stress from schoolwork factor that draws students to submit AI generated assignments, leading to a natural decrease in students' optimal use amounts.

#### *Elasticity of AI submission w.r.t. genuine learning*

AI, when used as a complement, is perceived to increase genuine learning; however, when used as a substitute, it is shown to decrease genuine learning (Chirikov, 2026). Since the student's choice variable represents the proportion of AI generated work submitted, AI use in this model reflects substitution away from genuine learning. Regardless of short term benefits and costs, the use of AI undermines deeper understanding necessary for long term intellectual benefit, showing that use ultimately has an inverse relationship with genuine learning (Schei et al., 2024).

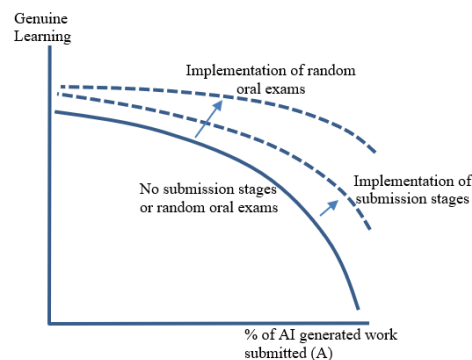


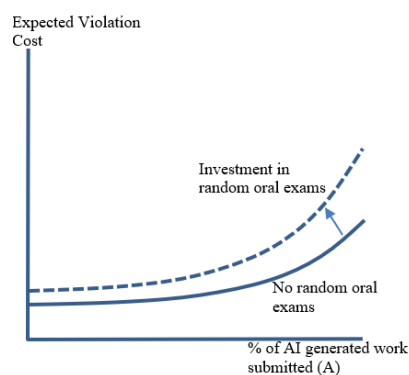
Figure 3: Impact of AI overreliance on the marginal utility of genuine learning.

When learning is highly valued, the marginal benefit of AI use decreases because of loss in long-term human capital, as AI is directly replacing human thought and work for the assignment. The first few AI generated assignments submitted will influence genuine learning the least, as the student is still putting effort into the rest of the assignments; however, as the proportion increases, genuine learning will drop at an increasing rate, reflecting the lack of human thought being put into schoolwork. If a student submits 100% AI generated assignments, they will have no genuine learning, as all of the work and thought has been offloaded to the AI.

Submission Stages increase genuine learning indirectly by requiring continued engagement and attention on a project, reducing last-minute completion and rotating the curve slightly up. However, there is still a chance students could submit AI generated work for each stage and make AI create output that could still track version histories, potentially lowering the magnitude of this relationship. Random Oral Exams increase genuine learning directly by requiring students to fully engage and internalize a deep understanding of their submitted work in preparation to verbally explain. Because students must be prepared for a potential oral exam, their genuine learning increases even at levels of no AI use and remains high, leading to the curve being shifted and rotated up.

#### *Elasticity of AI submission w.r.t. expected violation cost*

Many current policies rely on consequences to disincentivize AI use by raising the expected violation cost. The elasticity of AI submission with respect to consequences captures the responsiveness of student behavior to changes in expected violation cost.



*Figure 4: Sensitivity of AI reliance to the perceived marginal cost of academic violations.*

A large elasticity would suggest that policies punishing AI use are highly effective in lowering students' use patterns. Literature has suggested that this elasticity may be lower than expected due to ambiguity in policies and generally low perceived probability of punishment (Azevedo et al., 2026). Additionally, present bias overweighs current convenience over future consequences, which further reduces the elasticity. The random oral exam policy causes violations to become more salient and credible, which increases the expected violation cost and shifts the graph up slightly. As students submit higher proportions of AI generated work, the likelihood of having a random oral exam heightens and expected violation cost increasingly rises, which is reflected by an upward rotation of the curve. Students face shame when they are embarrassed by a lack of understanding in person. Without a policy in place, the anticipated shame is heavily discounted, as students are not thinking about the lack of learning from submitting AI generated work. The random oral exams policy increases the anticipated negative utility that arises, lowering the desire to submit AI generated work. However, present bias, ambiguity, and low perceived probability of punishment dampen the effect, suggesting that targeting enforcement alone may not be as effective as more heavily focusing on internal incentives.

## **Conclusion**

AI misuse and reliance have become increasingly prevalent in universities. This use creates tension between short term efficiency and long-term learning, which undermines the integrity of a college's degree. This study utilizes a unique game theory and principal-agent framework to illustrate the misaligned incentives between students and administrators. Students' AI use is highly influenced by internal factors such as stress from schoolwork and desire for genuine learning, indicating that policies targeting these factors can effectively reduce AI misuse.

Future research can explore long-term impacts of repetitive AI use, particularly examining how the elasticities of desire for genuine learning with respect to the amount of AI work submitted differs in the long term and the short term. Factors like repeated behaviors and skill atrophy could lead to different elasticities over long periods of time. Additionally, research could look into heterogeneity in behaviors among students and could examine how elasticities differ between areas of study, year, or general

academic performance. Future work could also explore behaviors and incentives around AI use in general. This work does not capture effects from complementary features of AI, which could uncover interesting insights.

To reduce students' optimal amount of AI generated work submitted, policies should target highly responsive incentives such as stress from schoolwork, efficiency, and desire for genuine learning. Incorporating transparency and clarity can provide a more sustainable approach to targeting AI misuse. Understanding elasticities and internal incentives can contribute to reducing misuse of AI while preserving genuine learning and degree integrity of institutions.

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# Prof. Hodgson's Notes

- Overall, I cannot tell you how impressed I am with both your writing and the quality of your analysis and your use of concepts from the major and the class! There are a couple of adjustments in your modelling I think you should think more about between now & the final draft. And your RoL should be a little more connected to your policies. That will be your biggest focus.

- What a great hook in your opening sentence!

- Your introduction was just plain stellar writing. I cannot tell you how impressed I am with the organization, rhetoric and clarity!

- What is the Unified Theory of Acceptance and Use of Technology model? When you referenced it on p. 4, I wasn't sure if this was something in the literature, and if so what type of thing.

- Similarly I wasn't sure what type of model AI-ICE was.

- Completed!!

- You have a really good use of both econ concepts and behavioral econ concepts, and the writing is really good.

- I love your use of information asymmetry and moral hazard, but I think you need to articulate more clearly (especially for moral hazard) how exactly this works in this context.

- Completed!

- The Tragedy of the Commons discussion on p. 6 could be fleshed out a little bit. It seems like there will be some contexts where professors rate AI work of higher quality and other contexts where this won't be the case. What are those?

- Added caveat of more creative assignments where AI may not excel – could expand more

- I don't have a clear vision of what you have in mind when you talk about "targeting social norms" on p. 6.

- Explained more!

- Present bias was a focus in the introduction, but doesn't appear much in the RoL. It needs at least a paragraph or two, if it is a highlight for you.

- Added!!

- I love your game theory, but it feels incomplete. You might add another table with variables instead of numbers and indicators of what a policy would need to achieve to be stable.

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- Having finished the RoL and game theory, this is fantastic background, but doesn't track your policies very closely. There needs to be more regular ties back to the debate between the two.

- I tried to directly tie in policies to the different aspects I talked about in the RoL

- **Also, with your game theory**, your policy that involves tracking the student's progress over time – that seems like it could be eventually gamed by AI that can type out things at any pace asked for. This might be a good discussion for the game theory section.

- I love your principal model. It captures exactly what it needs to capture and lays out the dilemma of administration with such clarity!

- Your efficiency variable – can you add more descriptors here. Efficiency of what? I don't have a picture in my mind.
  - Updating to "efficiency of completing schoolwork" → we can maybe clarify a bit more in the comparative statics section?
- The "Expected Violation Cost" in theory captures what is going on with the random orals model. However, I might argue that it doesn't do a good job of conveying the real psychological drivers here. There is something about the relationship with the professor, and/or the shame of showing you don't understand in an in-person setting that feels abstracted away by this framing.
  - Added to comparative statics for expected violation cost
- I very much appreciate genuine learning appearing in your model.
- I appreciate the focus on trade-off in your writing about the P-A models.
- The real cost of oral exams doesn't seem to be included here. This is cost of the professor's time and possibly something else. I'm not saying you necessarily need a 3rd layer in between (professor model), but that feels missing. The same could be said of the other policy, which might (?) require comparison of drafts?
  - 3 level model added!
- Your comparative statics are good (graphs), and the writing is OK. But I don't have a clear picture of this notion of efficiency as it might be thought of or experienced by the student.
- Give your figures labels (Figure 1)
  - Completed! But we should maybe reword so there is not repetition between the section headers and fig captions (I left a comment about this)
- **The shape** of the figure on p. 12 doesn't make sense to me. Stress is reduced whether any AI use is present? The writing didn't help me resolve my confusion.
  - I tried to explain better that the submission stages breaks it down so each submission is less stress, but will return to this.
- You do a nice job of talking about marginal cost & marginal benefit! The best I've seen!
- The graph on p. 13 makes a lot of sense to me and the writing compliments it.
- Your discussion of the graph on p. 14 convinced me the ambiguity is playing a big role, and I appreciated your connection with the literature. However, I do think there could be more going on here psychologically, and think you could adjust this part between now and the final draft to explore this.
  - Trying to add anticipated shame to the discussion
- Your intro seemed to imply that time discounting would be an important part of your analysis, but I don't think that ended up happening.